Huawei Access Node QoS Congestion Event (Microbursts, interface tail drops, QoS queue starvation) Troubleshooting SOP for JAY-ACC-35

Node: JAY-ACC-35 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_6643

Description

The Huawei access node, specifically JAY-ACC-35, running as an access role at the Jayanagar Metro Site, is susceptible to QoS congestion events, including microbursts, interface tail drops, and QoS queue starvation. This phenomenon arises due to the inherent characteristics of Huawei's QoS (Quality of Service) implementation, which is based on the Multi-Protocol Label Switching (MPLS) architecture. In MPLS, labels are used to identify and prioritize different types of traffic. However, when the QoS queue is subject to excessive congestion, the system's inability to effectively manage the label swapping process can lead to microbursts, interface tail drops, and QoS queue starvation. This can be exacerbated by factors such as oversubscription, high-traffic bursts, and insufficient QoS configuration. The result is a degradation of network performance, characterized by packet loss, increased latency, and jitter. Alarm ALARM_6643 is triggered when the system detects such a QoS congestion event.

Pre-requisite Checks

- Verify that the remote hands have been dispatched to the Jayanagar Metro Site.
- Check the status of the redundant link to ensure it is active and functioning properly.
- Confirm that the Huawei access node, JAY-ACC-35, is up and running without any other issues.

Potential Root Causes

Cause	Probability
Insufficient QoS Configuration	High
Oversubscription of Network Resources	Medium
High-Traffic Bursts	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos queue stats, Expected Output: Queue statistics: packets in queue, packets dropped, packets transmitted, Description: This command will display the QoS queue statistics, including the number of packets in the queue, packets dropped, and packets transmitted. This will help identify the root cause of the congestion event.

Command: display interface counters, Expected Output: Interface counters: packets received, packets transmitted, errors, Description: This command will display the interface counters, including the number
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of packets received, transmitted, and errors. This will help identify
any interface-related issues contributing to the congestion event.

Command: display mpls stats, Expected Output: MPLS statistics: labels
swapped, labels popped, packets transmitted, Description: This command
will display the MPLS statistics, including the number of labels
swapped, labels popped, and packets transmitted. This will help
identify any MPLS-related issues contributing to the congestion event.
```

Resolution Steps

Step 1: Step: Verify QoS Configuration, Command: display qos config, Description: Verify the QoS configuration to ensure that it is sufficient to handle the network traffic. Adjust the configuration as needed to prevent congestion events.

Step 2: Step: Adjust Network Resource Allocation, Command: display resource allocation, Description: Adjust the network resource allocation to prevent oversubscription of network resources. This may involve upgrading the network infrastructure or adjusting the resource allocation policies.

Step 3: Step: Implement Traffic Shaping, Command: configure traffic shaping, Description: Implement traffic shaping to prevent high-traffic bursts that can exacerbate congestion events. This may involve configuring traffic policing or shaping policies.

Post-Check Verification

```
Command: display qos queue stats, Expected Output: Queue statistics:
packets in queue, packets dropped, packets transmitted, Description:
Verify that the QoS queue statistics show a significant reduction in
packets dropped and a corresponding increase in packets transmitted.

Command: display interface counters, Expected Output: Interface
counters: packets received, packets transmitted, errors, Description:
Verify that the interface counters show a significant reduction in
errors and a corresponding increase in packets transmitted.

Command: display mpls stats, Expected Output: MPLS statistics: labels
swapped, labels popped, packets transmitted, Description: Verify that
the MPLS statistics show a significant reduction in labels popped and
a corresponding increase in packets transmitted.
```

Roll-back Procedures

- Step: Restore Original QoS Configuration, Command: undo configure qos, Description: Restore the original QoS configuration to its previous state.

- Step: Restore Original Network Resource Allocation, Command: undo configure resource allocation, Description: Restore the original network resource allocation to its previous state.

- Step: Restore Original Traffic Shaping Configuration, Command: undo configure traffic shaping, Description: Restore the original traffic shaping configuration to its previous state.